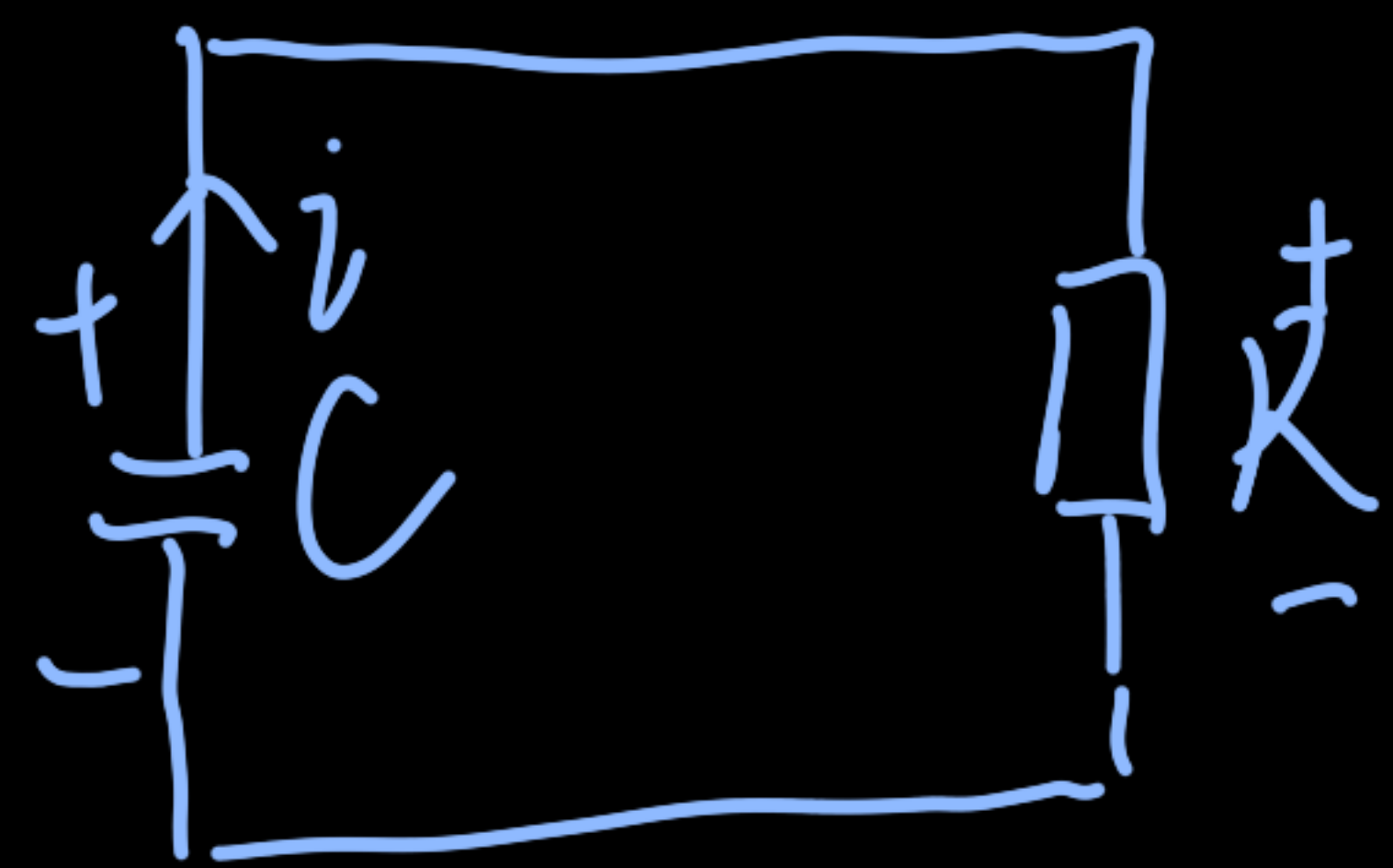


RC 的零输入方程



已知 $V_C(0_-) = V_0$

$$\begin{cases} V_C - V_R = 0 \\ i = C \frac{dv}{dt} \\ V_R = Ri \end{cases}$$

$$V_0 + RC \frac{dv}{dt} = 0$$

设 $\frac{dv}{dt} = p$

特征方程 $RCp + 1 = 0$

$p = -\frac{1}{RC}$ 特征根

则 $V_C = Ae^{-\frac{1}{RC}t}$

$$V_C(0_+) = V_C(0_-) = V_0$$

$$A = V_0$$

$$i = \frac{V_C}{R} = \frac{V_0}{R} e^{-\frac{t}{RC}} = I_0 e^{-\frac{t}{RC}}$$

$$i = C \frac{dV_C}{dt} = -C \cdot V_0 \left(-\frac{1}{RC}\right) = \frac{V_0}{R_0} e^{-\frac{t}{RC}}$$

齐次方程原理

$$RC \frac{dv_0}{dt} + V_C = 0$$

$$\frac{dV_C}{dt} = -\frac{1}{RC} \cdot V_C$$

$$\frac{d(e^{pt})}{dt} = p e^{pt}$$

假设 $V_C = A e^{pt}$

则 $\frac{dV_C}{dt} = A p e^{pt}$

$$A e^{pt} (RCp + 1) = 0$$

$$A e^{pt} \neq 0 \quad p = -\frac{1}{RC}$$

要恒成立..只

能使 $p = -\frac{1}{RC}$

则电路的状态